BYOUNGUK MIN

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RESEARCH INTERESTS

Computer architecture and compilers, with a focus on intermittent computing for battery-less energy-harvesting systems: GPUs and nonvolatile processors that stay correct and efficient under frequent power failures.

EDUCATION

Purdue University *Aug. 2021–Present*
Ph.D. in Computer Science (expected Dec. 2027)
Advisor: Prof. Changhee Jung

Yonsei University *Mar. 2019–Feb. 2021*
M.S. in Computer Science
Advisor: Prof. Bernd Burgstaller

Sungkyunkwan University *Mar. 2013–Feb. 2019*
B.S. in Computer Science

PUBLICATIONS

[1] **Byounguk Min**, Yilun Wu, Yuchen Zhou, Kasım Sinan Yıldırım, Khakim Akhunov, Jianping Zeng, Jongouk Choi, Wenjie Xiong, Dongyoon Lee, and Changhee Jung. "iGPU: Enabling GPUs for Intermittent AI on Batteryless Systems." In *Proceedings of the 59th IEEE/ACM International Symposium on Microarchitecture* (**MICRO'26**), Athens, Greece, Oct. 2026.

[2] Yida Zhang, Shao-Yu Huang, **Byounguk Min**, Andrew Inho Park, Gan Fang, Jongouk Choi, and Changhee Jung. "Achieving Speculative Intermittent Computation on Commodity Energy Harvesting Hardware." In *ACM SIGBED International Conference on Embedded Software* (**EMSOFT'26**), Oct. 2026.

[3] Noureldin Hassan, **Byounguk Min**, Changhee Jung, Yan Solihin, and Jongouk Choi. "WarmCache: Exploiting STT-RAM Cache for Low-Power Intermittent Systems." In *Proceedings of the 52nd ACM/IEEE International Symposium on Computer Architecture* (**ISCA'25**), Tokyo, Japan, Jun. 2025. [DOI]

[4] Yilun Wu, **Byounguk Min**, Mohannad Ismail, Wenjie Xiong, Changhee Jung, and Dongyoon Lee. "IntOS: Persistent Embedded Operating System and Language Support for Multi-threaded Intermittent Computing." In *Proceedings of the 18th USENIX Symposium on Operating Systems Design and Implementation* (**OSDI'24**), Santa Clara, CA, USA, Jul. 2024. [Link]

[5] Shinhyung Yang, Seongho Jeong, **Byounguk Min**, Yeonsoo Kim, Bernd Burgstaller, and Johann Blieberger. "Design-Space Evaluation for Non-Blocking Synchronization in Ada: Lock Elision of Protected Objects, Concurrent Objects, and Low-Level Atomics." *Journal of Systems Architecture* (**JSA**), vol. 110, 101764, 2020. [DOI]

RESEARCH PROJECTS

Enabling Heterogeneous Computing Systems Purdue University
Supervised by Prof. Changhee Jung *Dec. 2025–Present*

- Developing a power-aware work-partitioning scheme that splits each inference task across compute engines according to the available energy budget, with idempotent recovery when work is reassigned after a power failure.

IC-SIM: Intermittent Computing Simulator Purdue University
Supervised by Prof. Changhee Jung *Nov. 2025–Present*

- Building IC-SIM, a simulator for energy-harvesting systems that models power failures, capacitor/battery dynamics, and checkpoint/restart behavior to study end-to-end forward progress under intermittent power.

iGPU: Enabling GPUs for Intermittent AI on Batteryless Systems Purdue University
Supervised by Prof. Changhee Jung *Nov. 2022–Aug. 2026*

- Designed a GPU architecture for intermittent AI inference on batteryless energy-harvesting systems, a domain where prior architecture work has focused on CPUs.
- First-author paper published at MICRO'26.

WarmCache: STT-RAM Cache for Intermittent Systems Purdue University
Supervised by Prof. Changhee Jung *Nov. 2024–Jun. 2025*

- Contributed to WarmCache, an STT-RAM cache design with relaxed non-volatility for power-hungry energy-harvesting systems; published at ISCA'25.

IntOS: Embedded OS for Energy-Harvesting Platforms Purdue University
Supervised by Prof. Changhee Jung *Oct. 2023–Mar. 2024*

- Contributed to IntOS, an embedded OS and language support for multi-threaded intermittent computing on battery-less platforms; published at OSDI'24.
- IntOS offers conventional threads and transactions, with automatic undo-logging of persistent objects in non-volatile memory.

PBQ: Persistent Lock-Free Queue for NVM Yonsei University
Supervised by Prof. Bernd Burgstaller *Jan. 2020–Jun. 2020*

- Designed PBQ, a persistent lock-free queue that uses batching to achieve both high throughput and durability on non-volatile memory.
- Evaluated on an Intel Optane DC persistent memory server and a 120-core manycore server.

Relaxed Memory Consistency Evaluation on RISC-V Yonsei University
Supervised by Prof. Bernd Burgstaller *Jan. 2020–Apr. 2020*

- Extended the Ada concurrent-object work to RISC-V, evaluating non-blocking concurrent objects built on its atomic read-modify-write operations.
- Measured scalability of non-blocking algorithms on a 4-core SiFive HiFive Unleashed board.

Non-Blocking Synchronization in the Ada Programming Language Yonsei University
Supervised by Prof. Bernd Burgstaller *Apr. 2019–Nov. 2019*

- Proposed concurrent objects, a language-level primitive that encapsulates non-blocking synchronization built on atomic read-modify-write operations under relaxed memory models.
- Evaluated on x86 and ARMv8 to characterize how memory consistency models affect performance; published in JSA (2020).

TEACHING EXPERIENCE

CS 35200: Compilers: Principles and Practice — Purdue University
Head TA (Fall 2026); TA (Fall 2023, 2024, 2025) — *Instructor: Prof. Changhee Jung*

· Undergraduate compilers course. As Head TA, coordinate the TA team, programming-assignment logistics, and course communications.

CS 50200: Compiling and Programming Systems — Purdue University
TA (Spring 2024, 2025, 2026) — *Instructor: Prof. Changhee Jung*

· Graduate course on advanced compiler design.

CSI4104: Compiler Design — Yonsei University
TA (Fall 2019) — *Instructor: Prof. Bernd Burgstaller*

· Programming language design patterns and compiler techniques.

CSI2100: Computer Programming — Yonsei University
TA (Spring 2019, 2020) — *Instructor: Prof. Bernd Burgstaller*

· Python programming fundamentals and programming techniques.

TECHNICAL SKILLS

Languages	C, C++, Python, Ada
Simulators	gem5, GPGPU-Sim / Accel-Sim
Compilers	LLVM compiler infrastructure
Systems	Parallel and concurrent programming; persistent-memory programming (PMDK)

MILITARY SERVICE

Republic of Korea Army, KATUSA — *Jun. 2015–Mar. 2017*
Korean Augmentation To the United States Army, serving with United States Forces Korea (USFK)